ENTERPRISE DATA GOVERNANCE FRAMEWORK IMPLEMENTATION FOR E-COMMERCE PLATFORM

DATA GOVERNANCE TARGET OPERATING MODEL (TOM)

1. EXECUTIVE SUMMARY

The Data Governance Target Operating Model (TOM) defines the structure, roles, processes, and technologies required to manage data as a strategic asset across the enterprise. It ensures alignment between business objectives, regulatory requirements, and data management practices.

2. OBJECTIVES

- Establish clear accountability for data ownership and stewardship
- Improve data quality and trust across all business domains
- Enable regulatory compliance and auditability
- Standardize data governance processes across the enterprise
- Support scalable governance aligned with modern data platforms

3. OPERATING MODEL OVERVIEW

3.1. Governance Model Type

✓ **Hybrid / Federated Governance Model**

- Central governance defines policies & standards
- Domain teams execute governance within business units

4. GOVERNANCE STRUCTURE

4.1. Organizational Layers

Executive Layer

- Chief Data Officer (CDO)
- Data Governance Council

Strategic Layer

- Data Governance Lead
- Domain Data Owners

Operational Layer

- Data Stewards
- Data Custodians (IT)

Supporting Functions

- Risk & Compliance
- Legal
- Security

4.2. Roles & Responsibilities

Role	Responsibility
Data Owner	Accountable for data domain quality & compliance
Data Steward	Ensures data quality, metadata, definitions
Data Custodian	Technical management of data
Governance Council	Decision-making & policy approval

5. RACI FRAMEWORK

Activity	CDO	DGO	Owner	Steward	Custodian
Policy Definition	A	R	C	I	I
Data Quality Monitoring	I	C	A	R	C
Metadata Management	I	C	A	R	C
Access Control	I	C	A	C	R
Issue Resolution	I	C	A	R	C

6. CORE GOVERNANCE PROCESSES

6.1. Data Quality Management

- Define critical data elements (CDE)
- Establish data quality rules & thresholds
- Monitor via dashboards
- Issue management workflow:
    - Detect → Log → Assign → Resolve → Validate → Close

6.2. Metadata Management

- Maintain enterprise data catalog
- Define business glossary
- Track data lineage (source → transformation → consumption)

6.3. Data Access & Security

- Role-Based Access Control (RBAC)
- Attribute-Based Access Control (ABAC) (optional advanced)
- Access request & approval workflow
- Periodic access review

6.4. Data Lifecycle Management

- Define lifecycle stages:
    - Create → Store → Use → Archive → Delete
- Retention policies based on regulatory requirements

6.5 Data Classification

- Define classification levels:
    - Public
    - Internal
    - Confidential
    - Sensitive (PII)

7. GOVERNANCE ARTEFACTS

Mandatory Artefacts:

- Data Governance Policy
- Data Standards
- Data Quality Rules
- Data Classification Policy
- Data Access Policy
- Data Dictionary / Glossary
- Data Lineage Documentation
- Governance Control Register

8. TECHNOLOGY & TOOLING

8.1. Data Governance Tools

- Data Catalog: Collibra / Alation / Atlan / Microsoft Purview
- Data Quality: Informatica / Great Expectations

8.2. Data Platforms

- Data Warehouse (Snowflake, BigQuery)
- Data Lake / Lakehouse (Databricks)

8.3. Integration

- Metadata ingestion from ETL pipelines
- Automated lineage tracking

9. INTEGRATION WITH DATA ARCHITECTURE

- Governance embedded in:
    - Data ingestion pipelines
    - Transformation layers
    - BI / reporting layer
- Controls applied at:
    - Data creation
    - Data storage
    - Data access

10. COMPLIANCE & RISK MANAGEMENT

- Align with regulatory frameworks (e.g., GDPR principles)
- Implement audit trails
- Periodic compliance assessment
- Risk identification and mitigation

11. KPIs & SUCCESS METRICS

Metric	Description
Data Quality Score	>95% of data meeting quality standards
Issue Resolution Time	Avg time to resolve data issues
Data Availability	100% uptime of critical datasets
Metadata Completeness	>98% of documented data assets

12. IMPLEMENTATION ROADMAP

Phase 1: Assessment

- Current state analysis
- Gap identification

Phase 2: Design

- Define TOM
- Develop policies & standards

Phase 3: Pilot

- Implement in selected domain

Phase 4: Rollout

- Scale across enterprise

Phase 5: Optimization

- Continuous improvement

13. CHANGE MANAGEMENT & ADOPTION

- Stakeholder engagement plan
- Training & awareness programs
- Governance champions per domain
- Communication strategy

14. ESCALATION FRAMEWORK

- Data issues escalated:
    - Steward → Owner → Governance Council
- SLA for resolution defined

15. REVIEW & CONTINUOUS IMPROVEMENT

- Quarterly governance review
- Annual TOM update
- Continuous KPI monitoring

16. APPENDICES

Appendix A: Data Quality Dimensions

- Accuracy
- Completeness
- Consistency
- Timeliness

Appendix B: Sample Workflow

- Data Issue Management Workflow Diagram

Visual Diagram (Architecture + Governance Flow)